# FloatSat
# Aerospace Inf. Tech.

*Welcome*

*To the satellite builders group*

Julius-Maximilians-
UNIVERSITÄT
WÜRZBURG

Informationstechnik für Luft- und Raumfahrt
Aerospace Information Technology

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Aerospace Information Technology

# AEROPSACE INFORMATION TECHNOLOGY

Sergio Montenegro

# Development Examples

1980'er (The First)

2011'er (The Last)

30 years in between

1x constructed… den use for 30 Years

# Development Examples

1980'er (The first)

2011'er (The last)

1958

2014

**60 Years development**

# Development Examples

Yesterday

**707: 1958**

Today

**787: 2014**

Tomorrow??

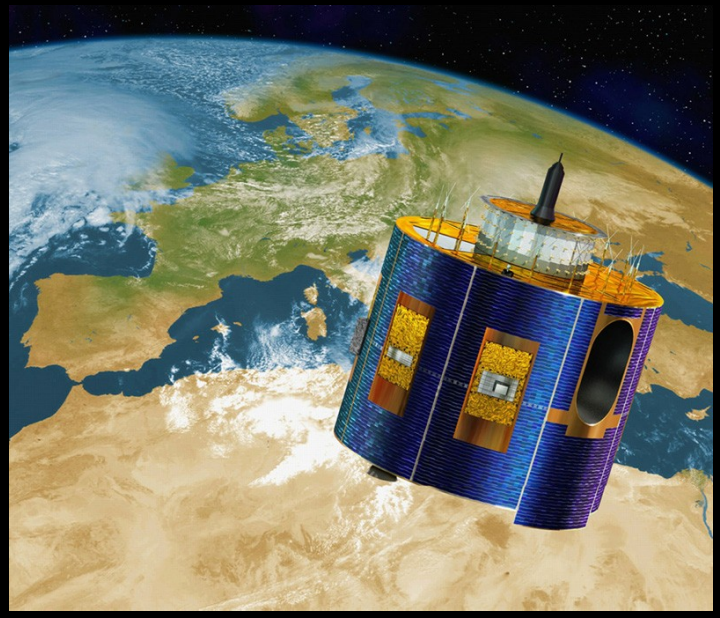
**202x???**

# Development Examples

Yesterday

**Tiros 1960**
**Meteorologisch**

Today

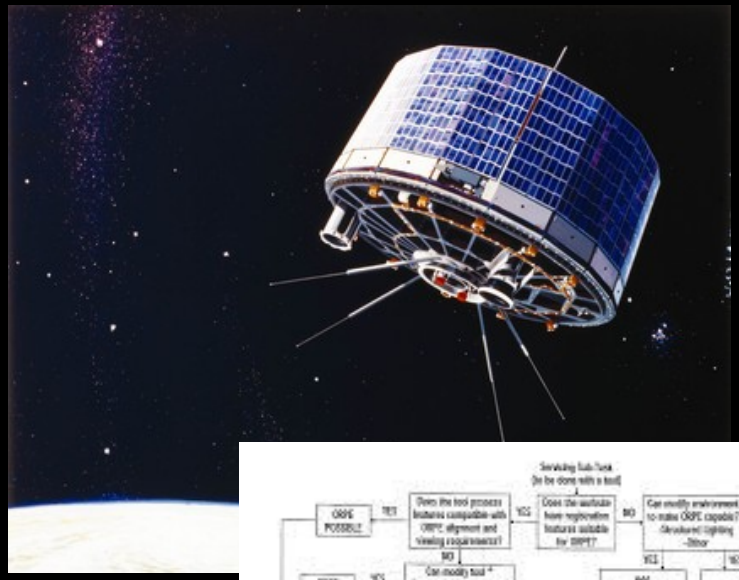


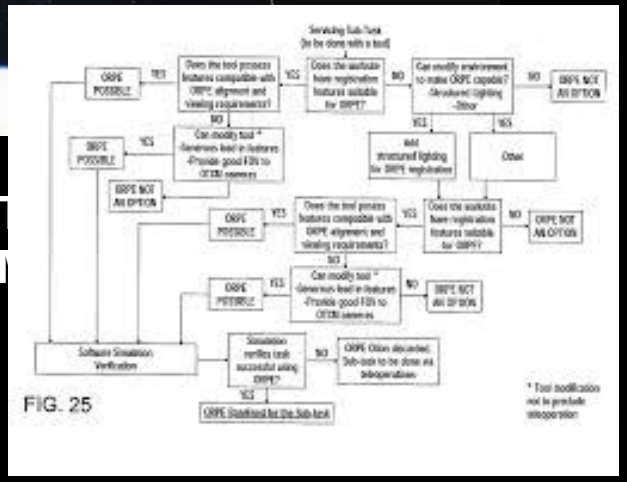
**Meteosat: 2013**

**53 years development**

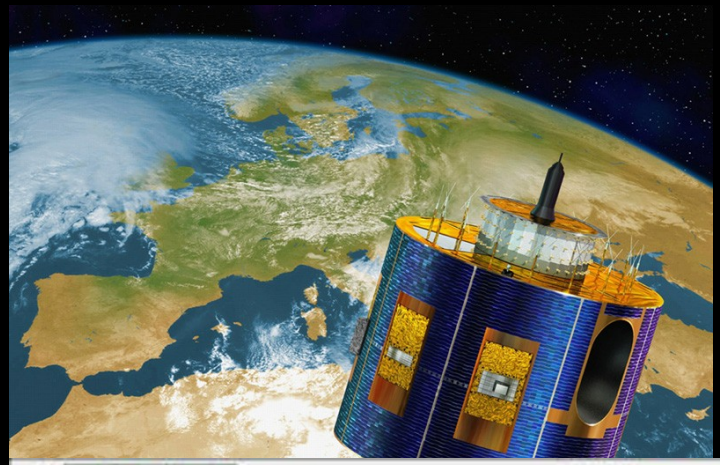
# Development Examples

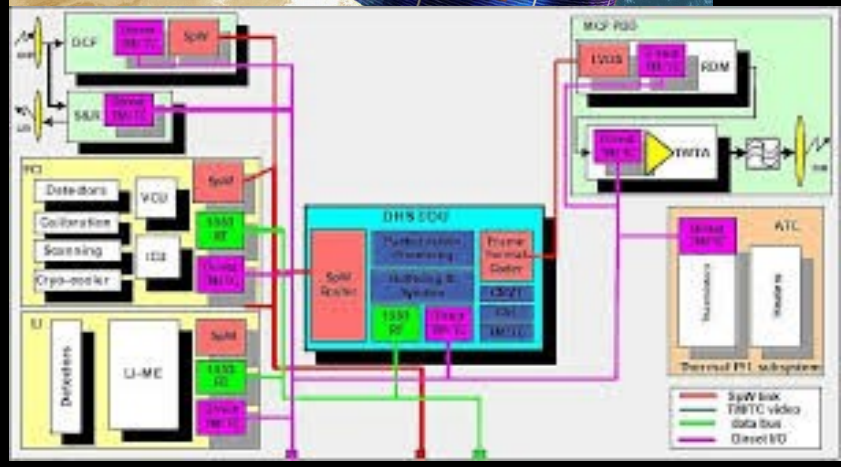
Yesterday






Tomorrow





# Our Research Emphasis

„Design to Survive“ Überlebensmaschinen
Ultra High dependability

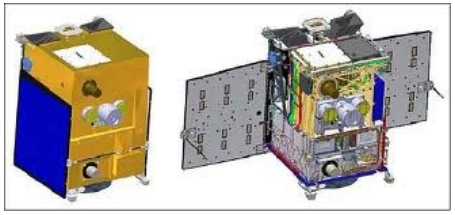
Autonomy,  adaptability

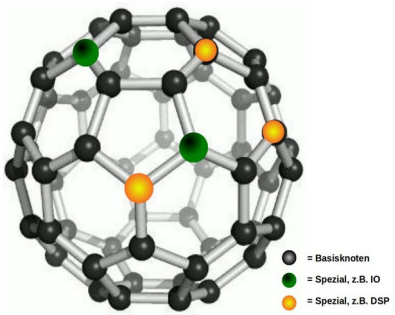
# How we do it

Waht is more impoortant?

| | |
|---|---|
| 1. Aovid crashes? | No! Let it crash! : Ultrafast Restart/Recovery |
| 2. Computer cetric? | Computer is not so important. The Nekwork is it! |
| 3. Mang. Complex Systems? | No! „Irreducible complexity“, Avoid complexity |
| 4. Robust? | No! Adaptability and intelligent diagnosis |

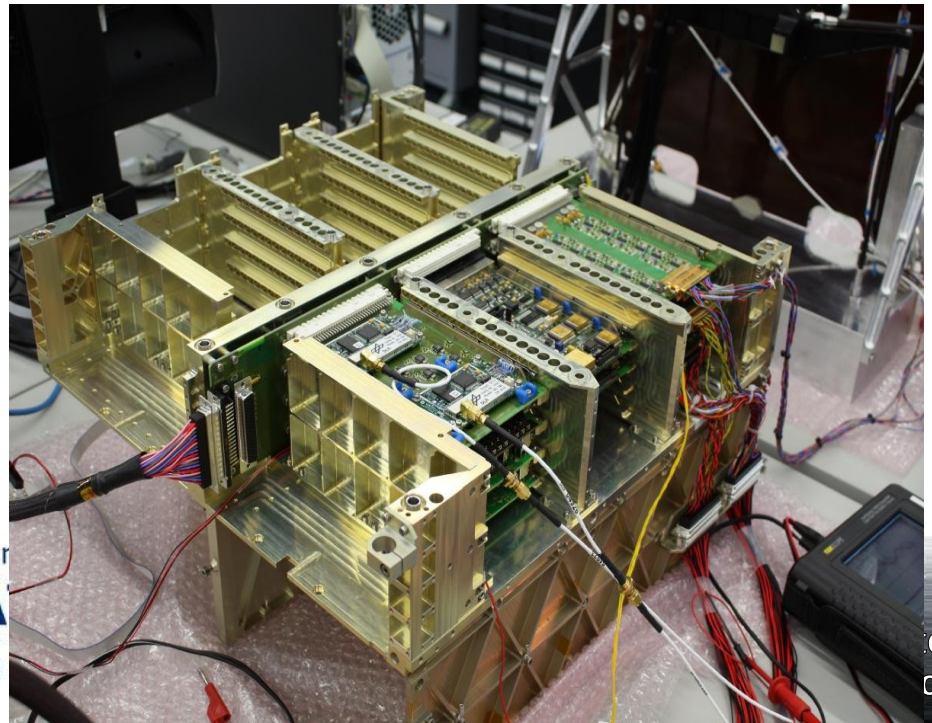
# Redundancy Management



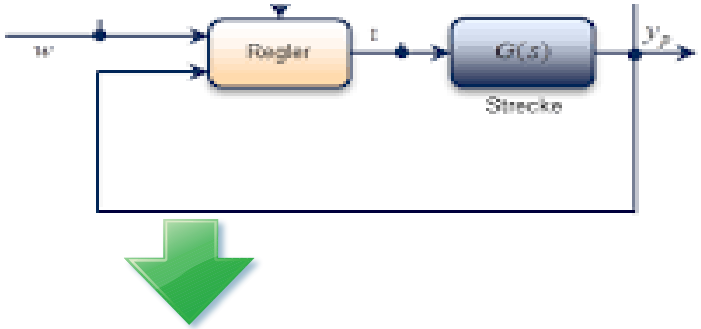


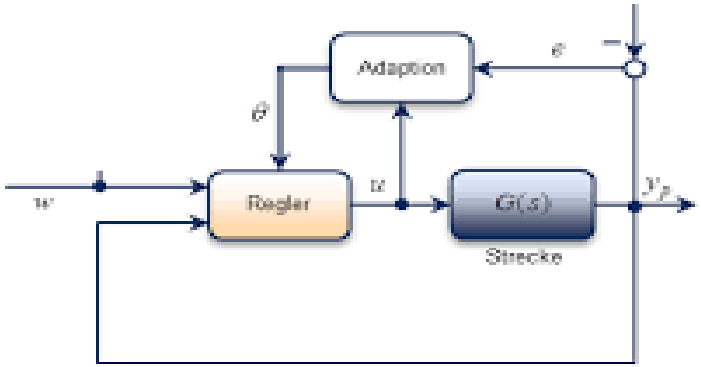

To Vidana

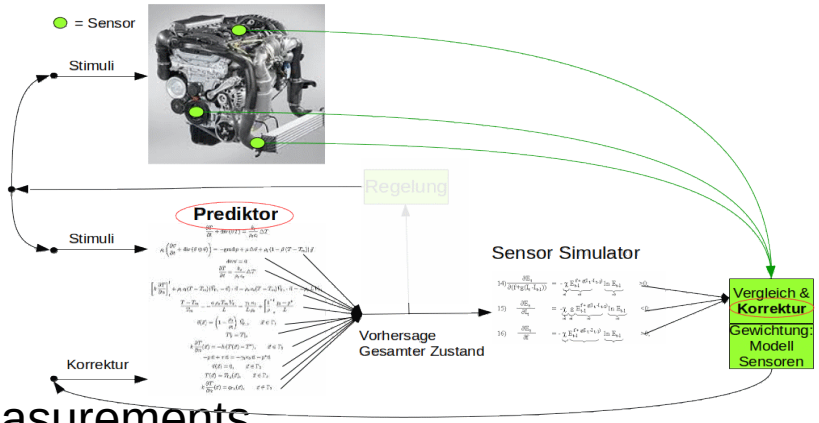
From Bird, TET, BIROS



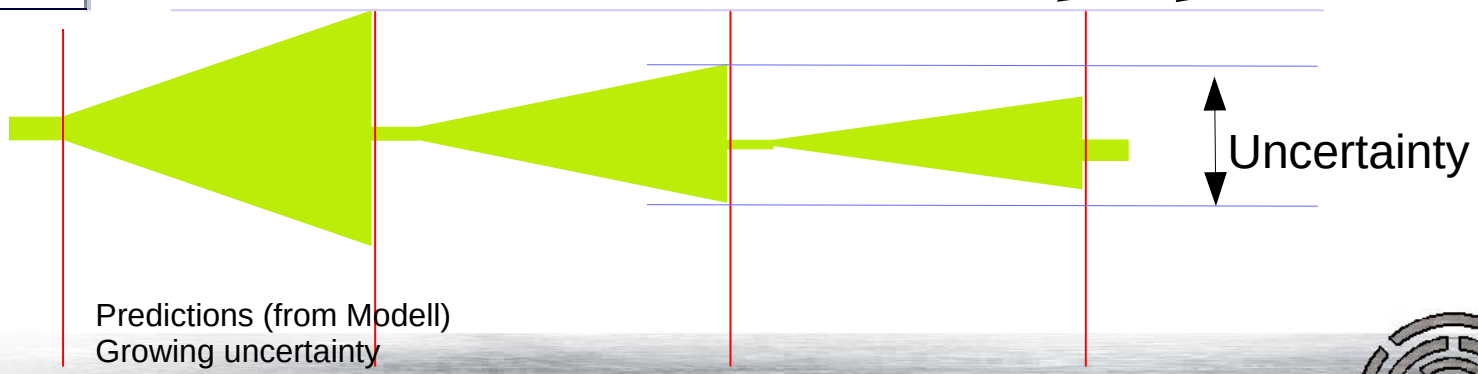
# Daptability



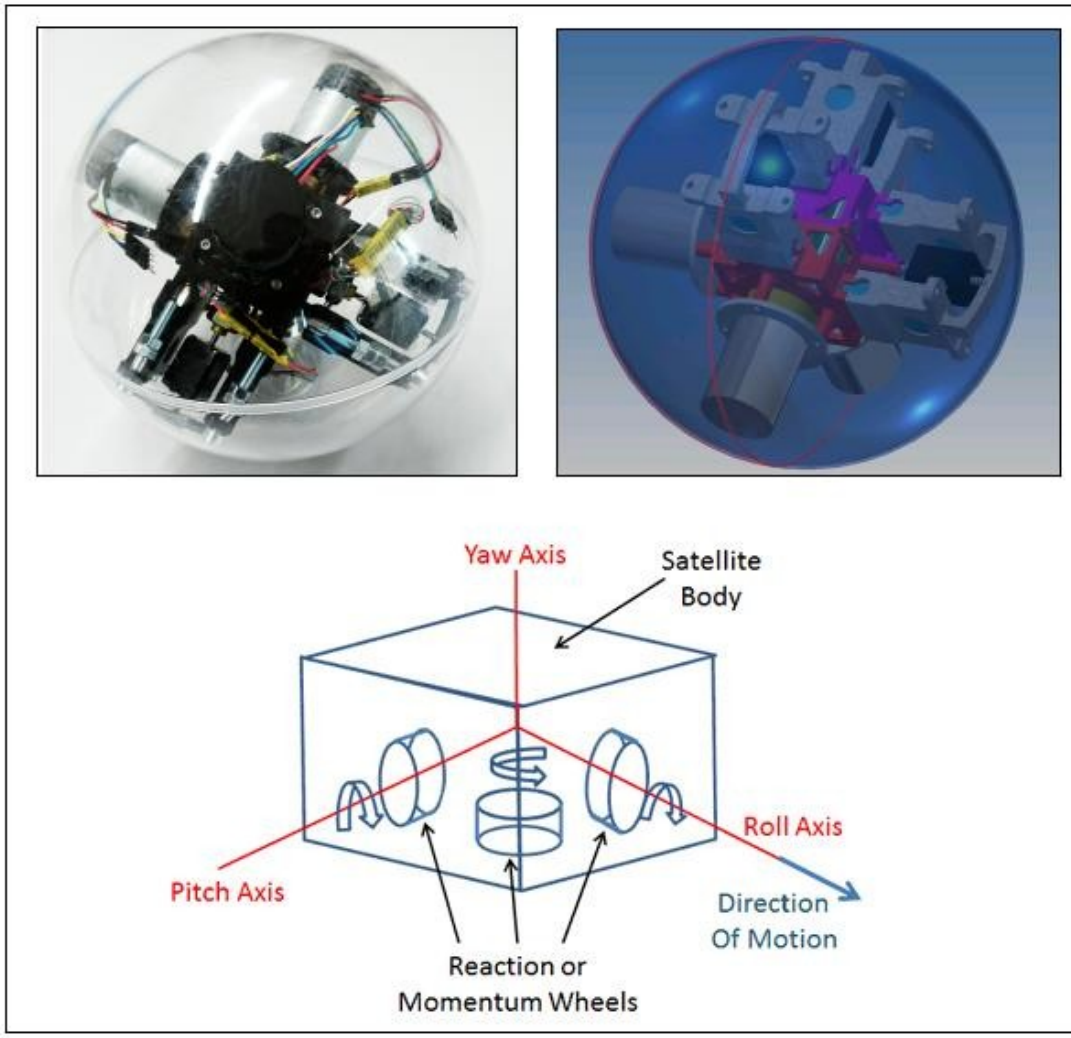

Measurements



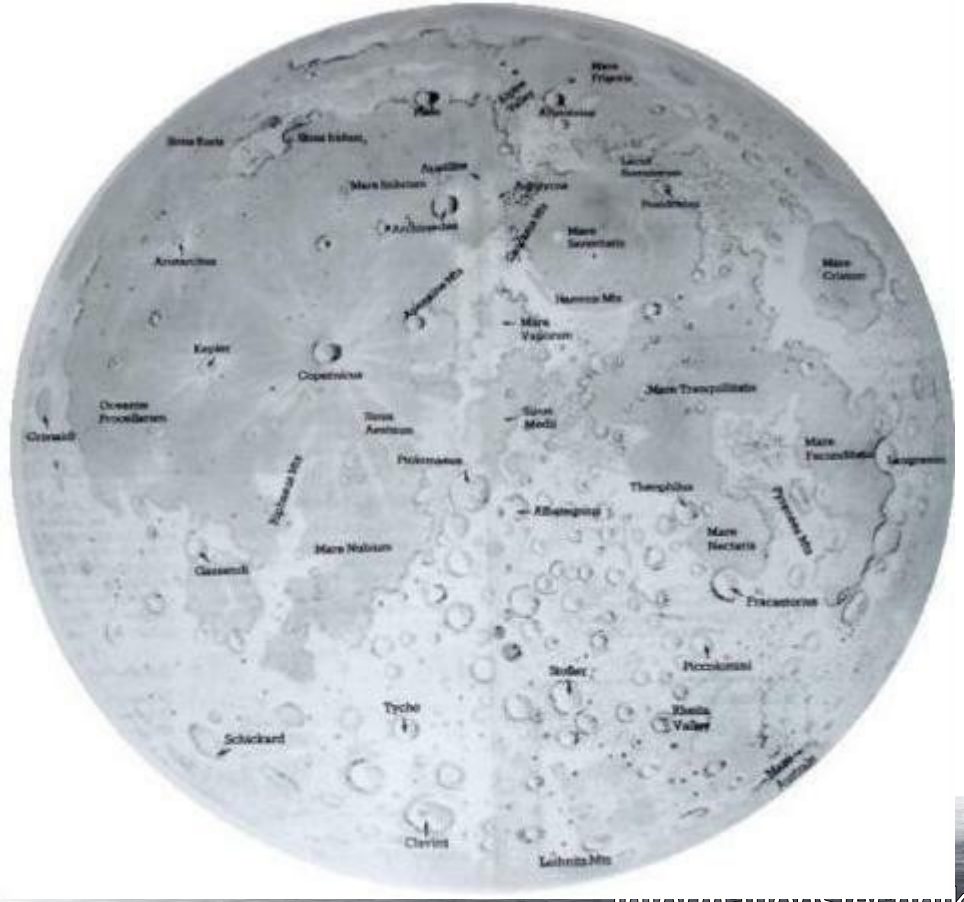

Predictions (from Modell)
Growing uncertainty

Play Midras

# Lerning Machines



# Lerning Machines




Optical Navigation above the moon

# Lerning Machines

SLAM
Simultaneous Localization
And Mapping

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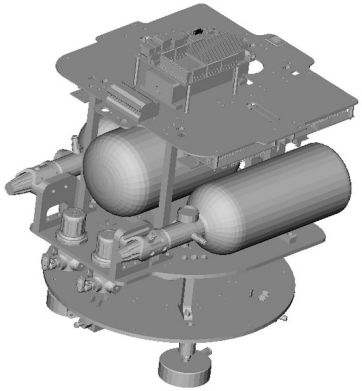
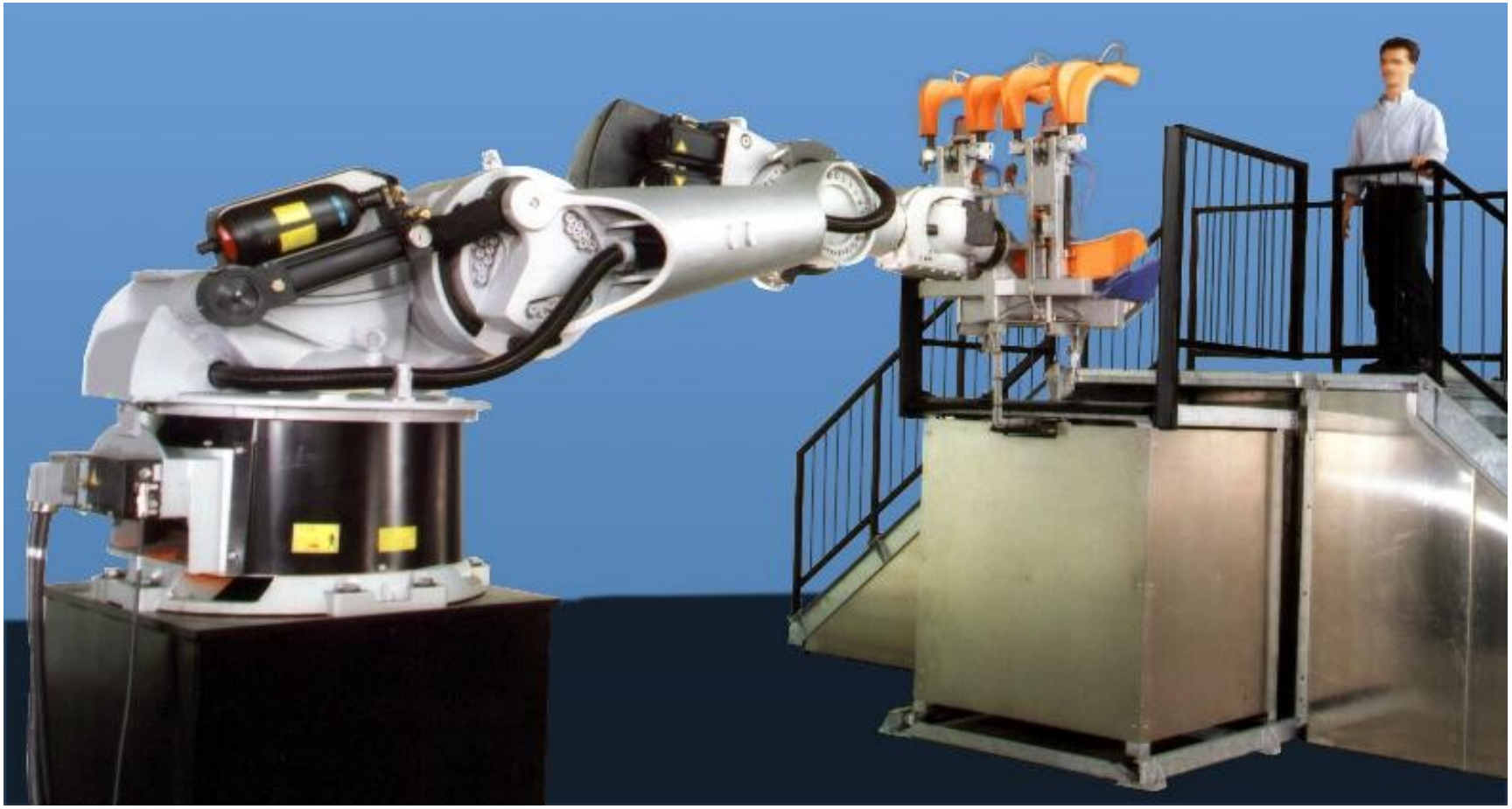
Informat
Aero

# Distributed Intelligence & Control

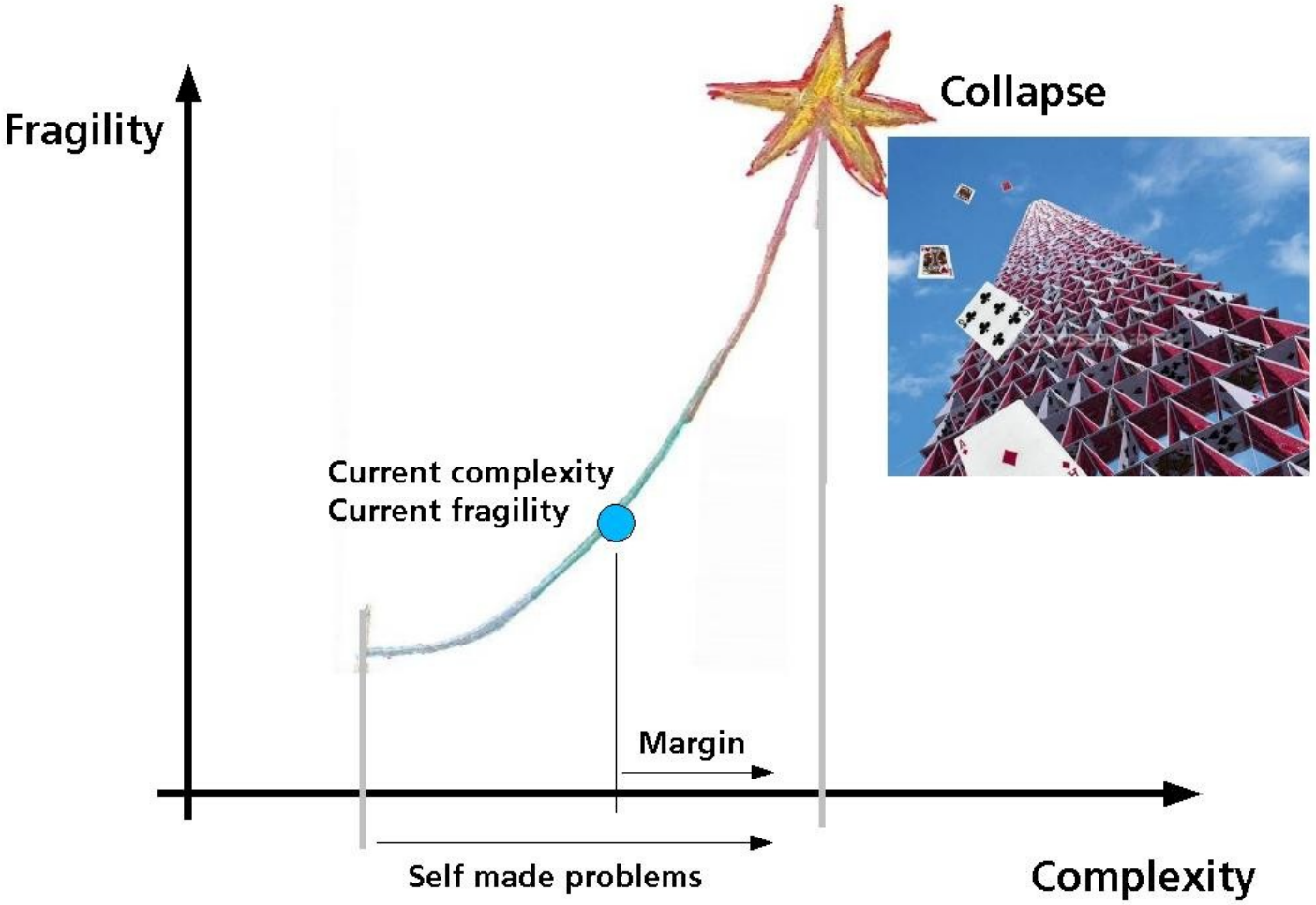


# Why complex systems are not dependable?

# Complexty and function

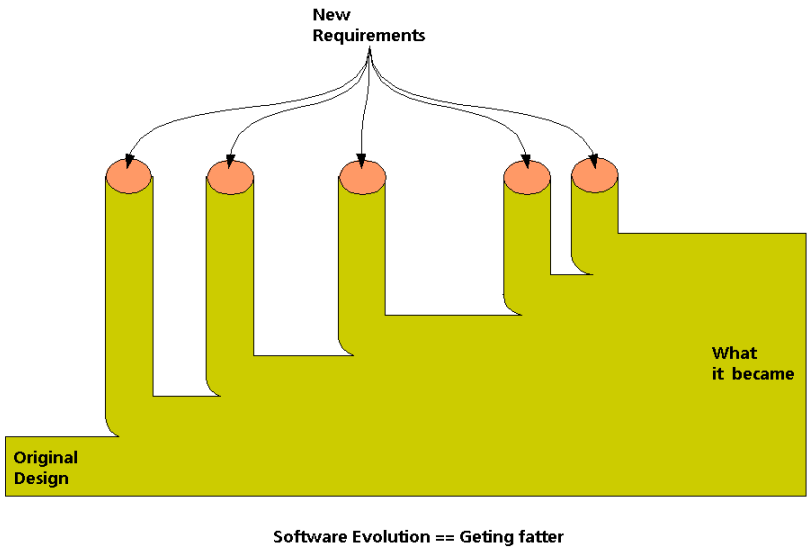


# Irreducible Complexity

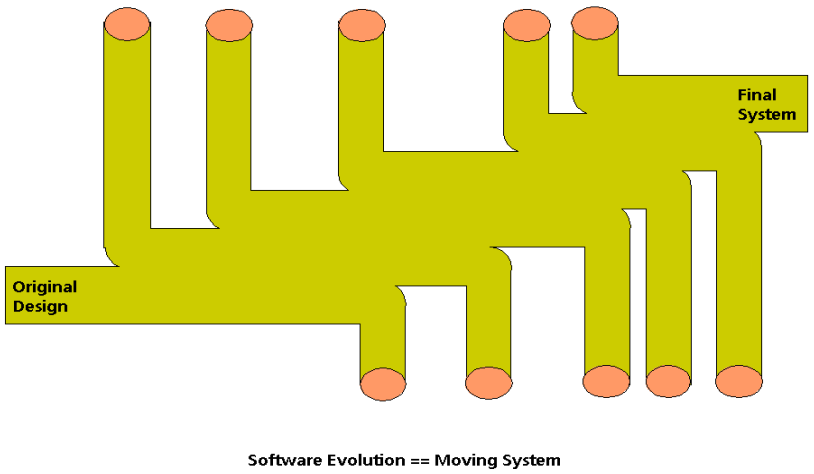


# Irreducible Complexity

## „Normal“




## What we do...



# Michelangelo,
# How did you du that?

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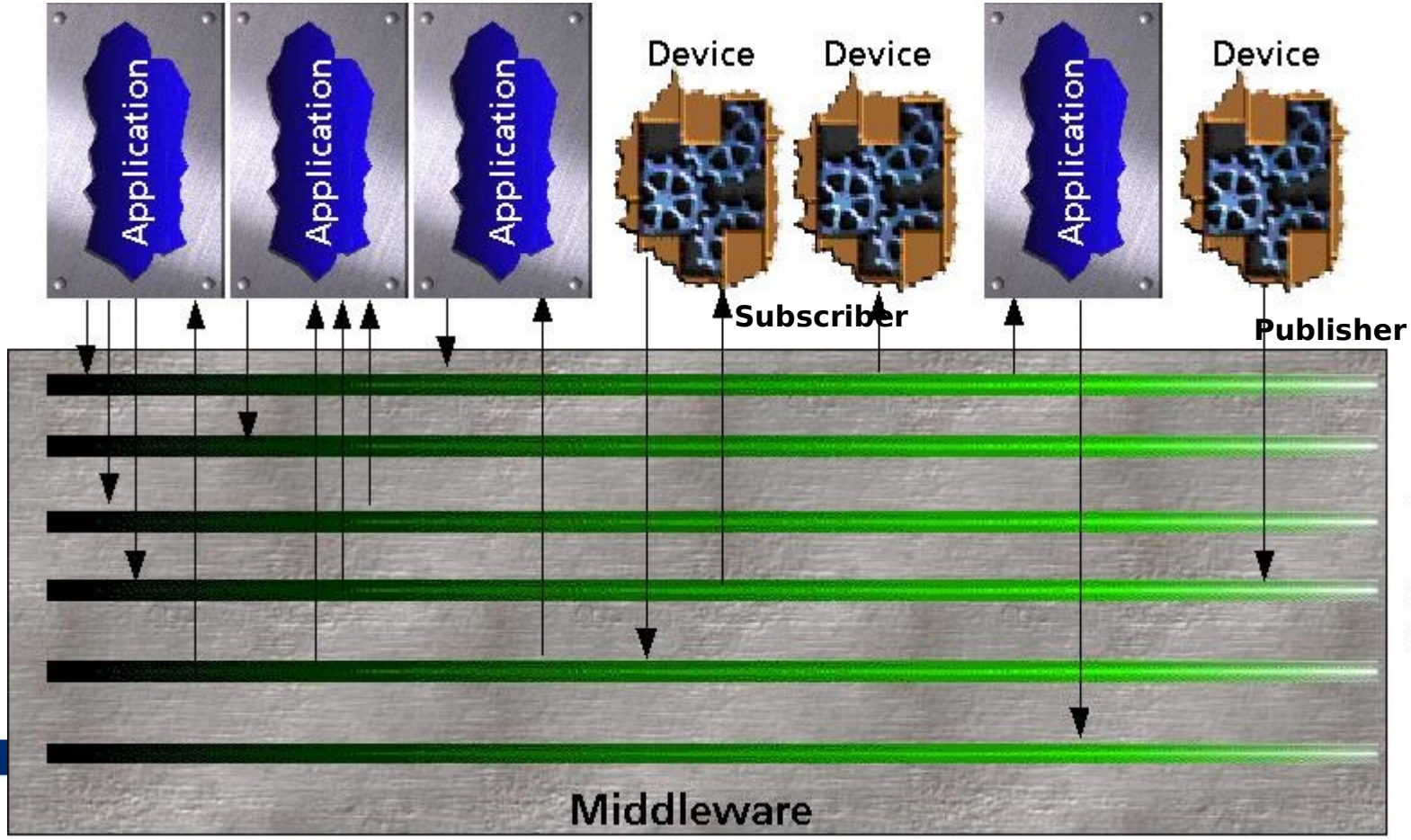
# It was already there!
# I just got rid of the excess

That is the art in
Software development

To find the minimum
and get ride of the excess

# Our Software Infrastructure

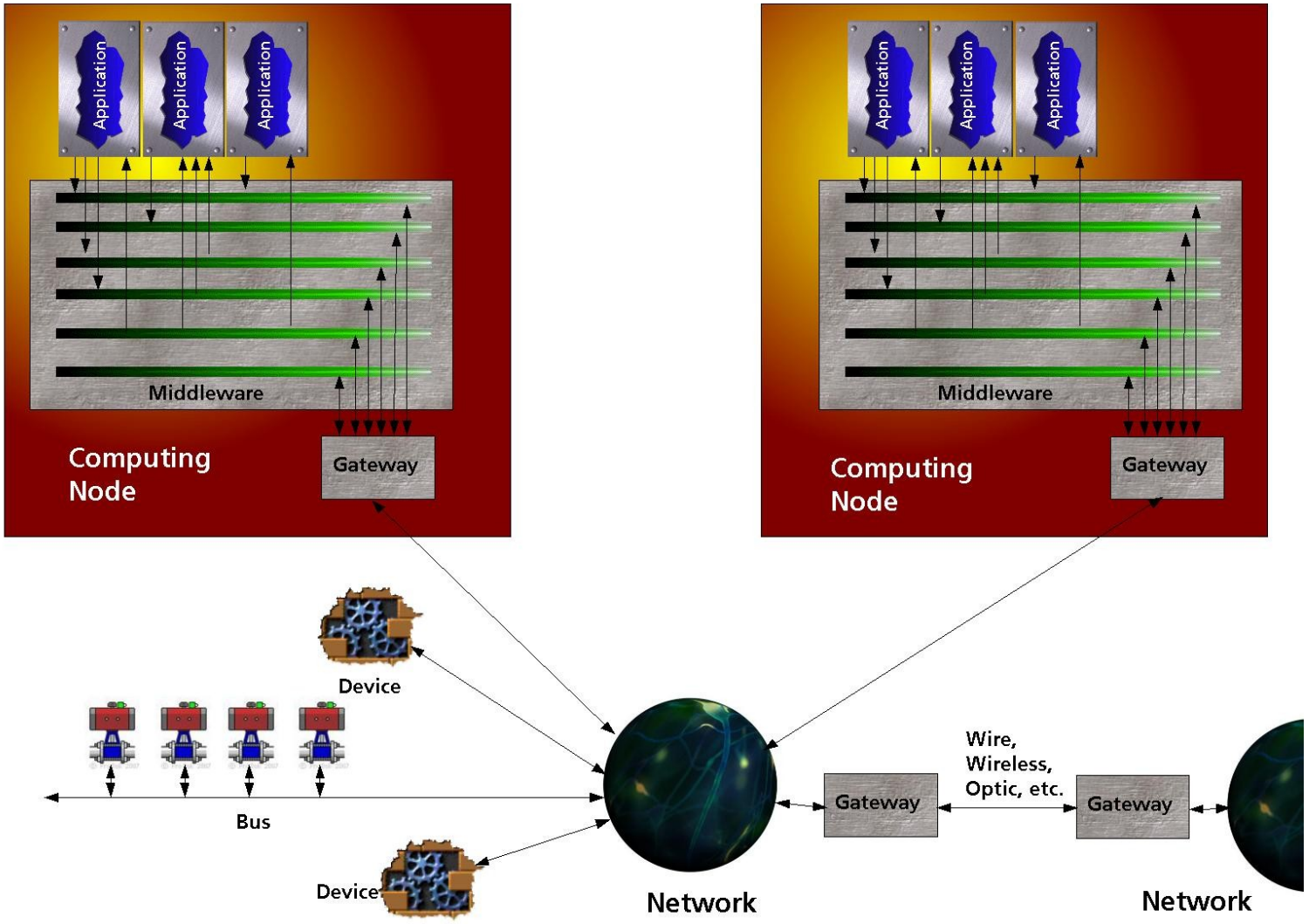



Connecting
SW ↔ SW
SW ↔ HW
HW ↔ HW

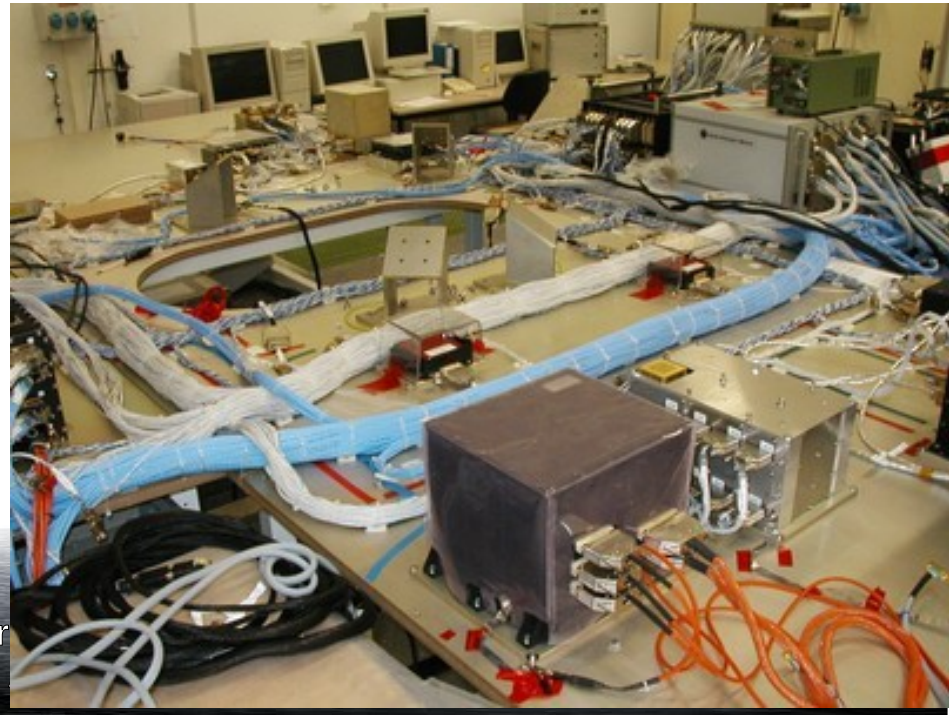
Topics ~=

Communication Channels

# Our Software Infrastructure

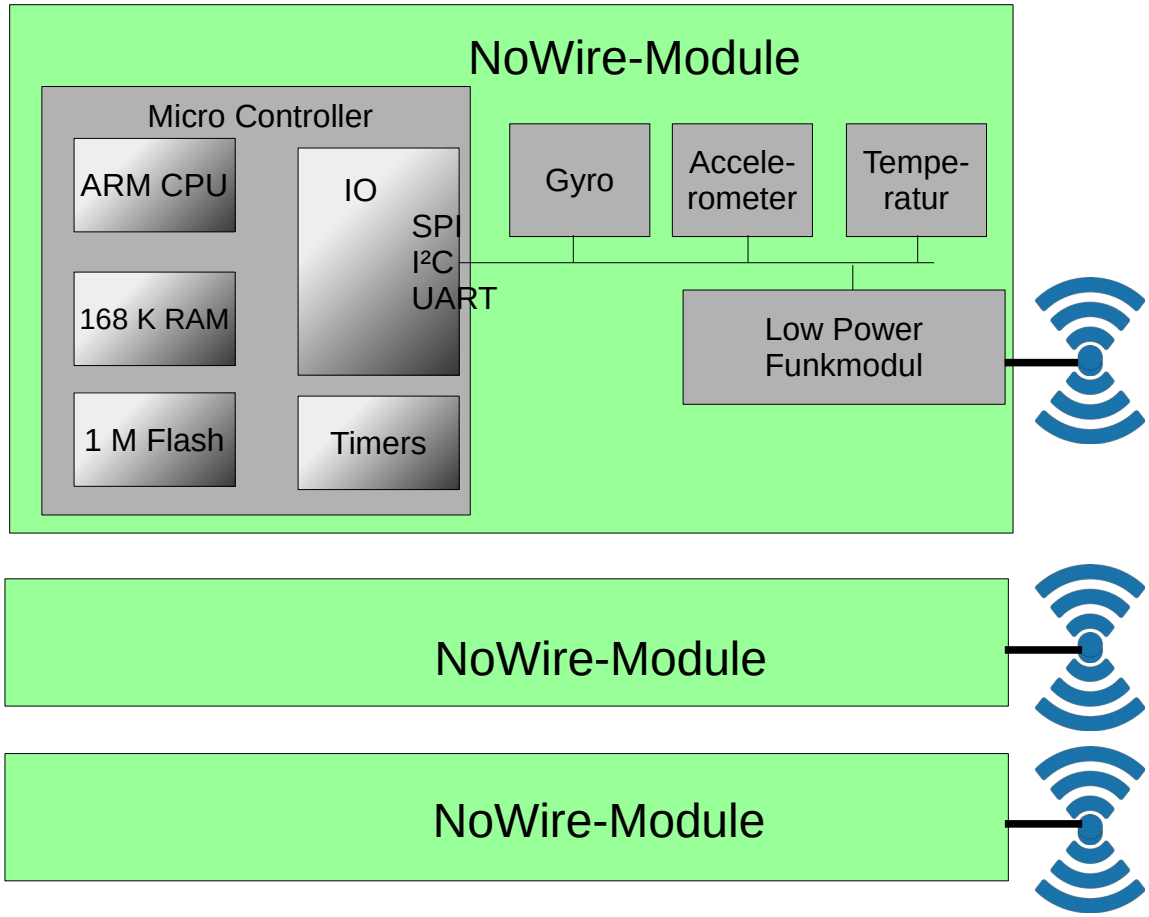


# Harness inside



# From hier ------→ To hier

# Skith




Boardcomputer ist unabhängig von IO Devices (Sensoren, Aktuatoren)
Maximale Flexibilität um Devices zu ersetzen / hinzuzufügen

# Other Projects

# Why Aerospace Information Technology

Why IT for Aerospace

Why an Extra Information Technology

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# Why Aerospace Information Technology

Why? IT: The most important component :)

Well the same says energy, orbit, structure, communications,....

(Who is right?)

Why computer?

control: Computer + ?

But: Same like on earth?

1 + 1 = ?

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# First question:
# Is space something special for the software?

Same like on earth?
1 + 1 = 2

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# First question:
# Is space something special for the software?

Same like on earth?
1 + 1 = (very) probably 2

Problems:
- **radiation**
- vibration,
- shock,
- thermal,
- vacuum,
- no gravity

is it a hardware problem?
it is a SW Problem!

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# Radiation: Biggest Problem for Software!
# (Did I Say Software? is it not Hardware?)

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# First question:
# Is space something special for the software?

1 + 1 = (very) probably 2
and the computer will probably work!
and the mission will probably succeed
(if you say this you are out)

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# First question:
# Is space something special for the software?

1 + 1 = (very) probably 2
but the computer will ~~probably~~ work!

we know, the Hardware will fail,
to survive in is the hands of the software

**Design to Survive!**

# We are searching for dependability

**Fault Tolerance + Low Prog. Error count + Adaptability (Autonomy)**

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# We are searching for dependability

10, 15, 20 Years working (alone) in Space
A 10 years Trip
Explore a planet for one year
…

… with no external help, no maintenace, no repair
if some thing goes wrong, rescue your self!

+ Adaptability + Autonomy

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Movies Supreices

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# Danke!

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